

Solution and scoring guide

Question 1 | 10 points total | Fictional sample: Farah Aziz

A complete proof

At $n = 1$, the sum is $1 = 1^2$. Let $k \geq 1$ be arbitrary, and assume $1 + 3 + \dots + (2k - 1) = k^2$. The next odd number is $2(k + 1) - 1 = 2k + 1$, so add this term to the sum in the hypothesis.

$$\begin{aligned} 1 + 3 + \dots + (2k - 1) + (2k + 1) \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

The base case and induction step hold. Therefore the identity holds for every integer $n \geq 1$ by mathematical induction.

Rubric and illustrative manual score: 4 / 10

Each criterion earns all of its points if met, or zero if not met. The sample earns:

Base case	2 / 2
Correctly verifies $n = 1$: both sides equal 1.	
Induction hypothesis	2 / 2
Correctly assumes the identity for arbitrary $k \geq 1$.	
Correct extension	0 / 3
Repeats $2k - 1$. The next term is $2k + 1$.	
Valid successor algebra	0 / 3
$k^2 + 2k - 1$ is not $(k + 1)^2$; it is 2 smaller.	

Feedback to guide a revision

Keep your base case and hypothesis. In the induction step, calculate the $(k + 1)$ st odd number using $2(k + 1) - 1$. Then expand $(k + 1)^2$ and compare its constant term with the expression on the previous line.

The 4/10 score is a manually checked reference. An automated result still requires instructor review before release.